

Frugal and responsible AI

What levers for data scientists?

Elise Pupier & Pierre-Loïc Bayart

Who are we ?

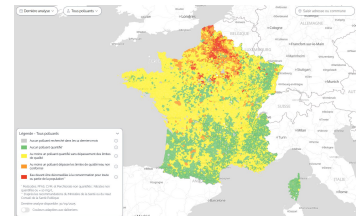
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*AI educational
research engineer*

Elise Pupier
Data analyst





- Community of **~2000 Tech volunteers** (Data, Dev, Designers) offering their skills for NGOs projects
- A plea to promote a more responsible and civic-minded digital world
- Each year : **10 to 20 projects** for the **environment, democracy or social justice**
 - **Pyronear**: Forest fire detection using video footage
 - **DansMonEau**: Interactive map of water quality by municipality
- **Local projects (Grenoble)**
 - **Mobilité Durable en Montagne** : Analyzing access to the mountains by public transport
 - **ChatVTT** :



**AI : what do you know
about its impacts ?**

Kahoot!

According to GreenIT, how many years of profitable reserves of critical metals do we have left?



< 20 years

References : <https://www.greenit.fr/2025/10/01/etat-des-reserves-mondiales-de-metaux/>

Electricity consumption in France : ~450 TWh/year.
What is the trajectory of global data center
consumption by 2030?



1500 TWh/year

References : <https://theshiftproject.org/app/uploads/2025/09/Synthese-RF-PIA-1.pdf>

A data center is planned for Eybens. What is the target energy consumption? (1 GW : a nuclear reactor)



200 MW

(this represents ~ 60% of the total average electricity consumption of Grenoble Alpes Métropole)

References : <https://www.placegrenet.fr/2026/01/05/datacenters-geants-a-eybens-et-villefontaine-entre-promesses-technologiques-et-inquietudes-des-populations-locales/667902>

What was the conflict between Anthropic and the Trump administration about in February?



The use of AI to monitor population

The use of AI to manufacture
autonomous weapons

Define the need

AI or not AI ? If AI, what type ?

Goal : "Predicting if a building's heating should turn on" Order from **Lowest** energy consumption of the model to **Highest**

Bon ordre :

Simple logic like "If $T < 17^{\circ}\text{C}$ then ON, if $T > 20^{\circ}\text{C}$ then OFF"

A logistic regression predicting heating needs from historical data.

A multi-layered neural network analyzing complex weather and sensor data

Asking a model like Claude to analyze the current thermal conditions

Define the need

Assess whether **AI** is actually **relevant**

*Is the problem complex enough?
Would simpler methods work?*

Clearly **define the business need** before thinking about AI

*What problem you're solving?
What functions are actually required?*

Consider **alternatives** to **(generative) AI**

*Could a simpler, non-generative approach achieve the same result?
What are the trade-offs compared to using generative AI?*

Evaluate **environmental impact** (full lifecycle)

*What is the total environmental cost of this solution over its entire lifecycle (not just training)?
Does a simpler alternative significantly reduce this impact?*

Define the **right level** of **performance**

*What level of performance is truly required to meet the user's need?
Are we over-optimizing beyond what brings real value?*



References :

[Kit d'engagement pour une intelligence artificielle frugale au sein d'une organisation](#), Ministères aménagement du territoire transition écologique, Ecolab

[Référentiel général pour l'IA frugale](#), AFNOR

[Intelligence artificielle, données, calculs : quelles infrastructures dans un monde décarboné ?](#), The Shift Project, Octobre 2025



Quantify the energy and environmental impact

CodeCarbon and Ecologits libraries

Order these energy sources from HIGHEST to LOWEST carbon intensity (kg CO₂/MWh)

Bon ordre :

Coal

Petroleum

Natural Gas

Hydroelectricity

Quantify the energy and environmental impact

Beyond classical metrics :

Data science metrics :

accuracy, precision, recall, f-score,
confusion matrix, ...

Business metrics :

churn, profit, losses, time, new
customers, ...



Environmental metrics :

Energy, Global Warming Potential, Abiotic Resource Depletion for
Elements, Primary Energy, Water Consumption Factor, ...

Quantify the energy and environmental impact

ML, DL, LLM...

On computer, server or
cloud VM

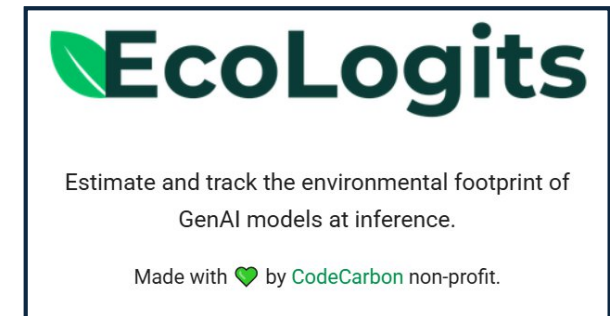
Generative AI
(LLM)

Estimation from API
inference calls



Sampling-based
(every 10 seconds)

Low overhead



Environmental impacts
per request

When tracking the environmental impact of training a machine learning model, what does CodeCarbon actually do?



It estimates hardware energy consumption and converts it into CO2 emissions

Quantify the energy and environmental impact

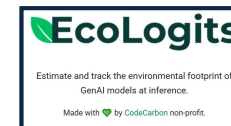
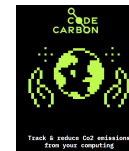
Use cases

Training a ML/DL model on a laptop

Running ML/DL pipelines in the cloud

Using LLM API in a web app

Library



Quantify the energy and environmental impact

⚠ Use CodeCarbon preferably on **Linux**, as it relies on accurate hardware detection



```
from codecarbon import EmissionsTracker
tracker = EmissionsTracker()
tracker.start()

# ML code

tracker.stop()
```

Initialization

Loads configuration

Detects system specifications (CPU, GPU, RAM)

Prepares logging and data storage

Quantify the energy and environmental impact



```
from codecarbon import EmissionsTracker
tracker = EmissionsTracker()
tracker.start()

# ML code

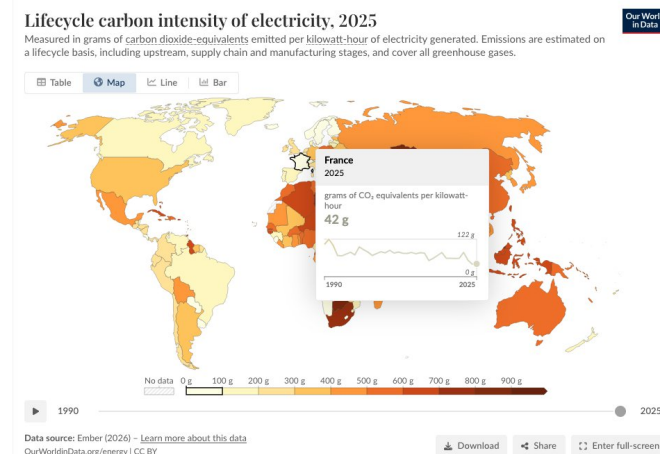
tracker.stop()
```

Hardware monitoring loop

Measures resource usage

Estimates power consumption

Applies regional carbon intensity



Quantify the energy and environmental impact



```
from codecarbon import EmissionsTracker
tracker = EmissionsTracker()
tracker.start()

# ML code

tracker.stop()
```

Stop tracking

Computes : energy consumed (kWh), CO₂ emissions (kgCO₂eq) and duration

Quantify the energy and environmental impact



Emissions: 0.0023 kg CO₂eq
Energy consumed: 0.005 kWh

Results generation

Console output

CSV log file

Monitoring tools
(CodeCarbon Dashboard, Prometheus,
cloud dashboards, ML flow...)

Quantify the energy and environmental impact

DL training loop on Google Colaboratory with GPU

Initialization

```
[codecarbon INFO @ 15:30:58] [setup] RAM Tracking...
[codecarbon INFO @ 15:30:58] [setup] CPU Tracking...
[codecarbon INFO @ 15:30:59] CPU Model on constant consumption mode:
Intel(R) Xeon(R) CPU @ 2.00GHz
[codecarbon INFO @ 15:30:59] [setup] GPU Tracking...
[codecarbon INFO @ 15:30:59] Tracking Nvidia GPUs via PyNVML
[codecarbon INFO @ 15:30:59] The below tracking methods have been set up:
    RAM Tracking Method: RAM power estimation model
    CPU Tracking Method: global constant
    GPU Tracking Method: pynvml

[codecarbon INFO @ 15:30:59] >>> Tracker's metadata:
[codecarbon INFO @ 15:30:59] Platform system: Linux-6.6.113+-x86_64-with-
glibc2.35
[codecarbon INFO @ 15:30:59] Python version: 3.12.13
[codecarbon INFO @ 15:30:59] CodeCarbon version: 3.2.6
[codecarbon INFO @ 15:30:59] Available RAM : 12.671 GB
[codecarbon INFO @ 15:30:59] CPU count: 2 thread(s) in 1 physical CPU(s)
[codecarbon INFO @ 15:30:59] CPU model: Intel(R) Xeon(R) CPU @
2.00GHz
[codecarbon INFO @ 15:30:59] GPU count: 1
[codecarbon INFO @ 15:30:59] GPU model: 1 x Tesla T4
[codecarbon INFO @ 15:30:59] Emissions data (if any) will be saved to file
/content/emissions.csv
```

Monitoring loop

```
Epoch 1/10
1875/1875 ————— 7s 2ms/step - accuracy: 0.9129 - loss: 0.2969
[codecarbon INFO @ 15:31:14] Energy consumed for RAM : 0.000042 kWh. RAM Power :
10.0 W
[codecarbon INFO @ 15:31:14] Delta energy consumed for CPU with constant : 0.000177
kWh, power : 42.5 W
[codecarbon INFO @ 15:31:14] Energy consumed for All CPU : 0.000177 kWh
Epoch 3/10
```

...

Results generation

```
[codecarbon INFO @ 15:31:51]
Graceful stopping: collecting and writing information.
Please wait a few seconds...
[codecarbon INFO @ 15:31:51] Energy consumed for RAM : 0.000143 kWh. RAM Power :
10.0 W
[codecarbon INFO @ 15:31:51] Delta energy consumed for CPU with constant : 0.000079
kWh, power : 42.5 W
[codecarbon INFO @ 15:31:51] Energy consumed for All CPU : 0.000610 kWh
[codecarbon INFO @ 15:31:51] Energy consumed for all GPUs : 0.000381 kWh. Total GPU
Power : 27.35822538416959 W
[codecarbon INFO @ 15:31:51] 0.001134 kWh of electricity and 0.000000 L of water were
used since the beginning.
[codecarbon INFO @ 15:31:51] Done!
```

Quantify the energy and environmental impact

LLMs : uncertainty in energy estimation

✗ Proprietary models

Model size unknown

Hardware not disclosed

No access to real
measurements



✓ Open-weight / open-source models

Model size is public

Can run locally

Measurable with physical devices



Start building with
open models

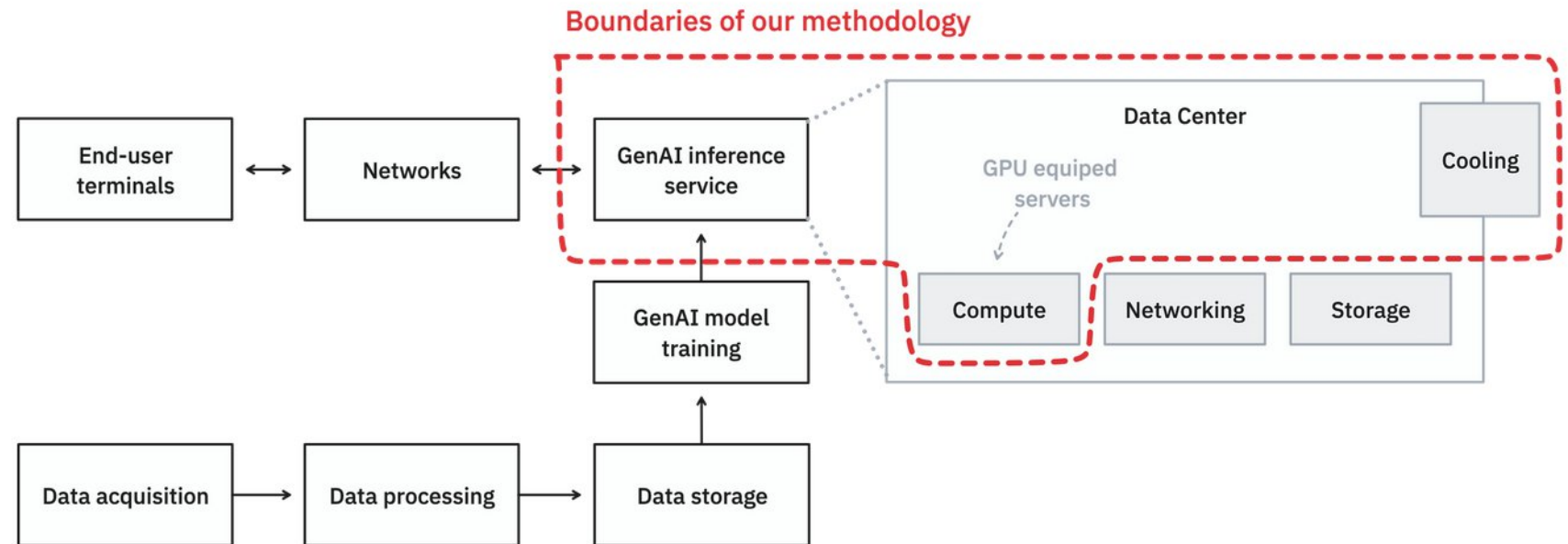
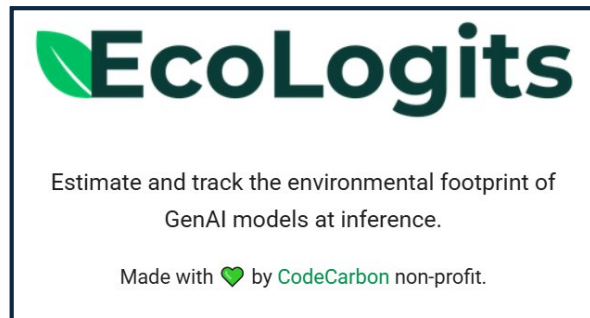


The **High-Throughput** and
Memory-Efficient inference and
serving engine for LLMs

Easy, fast, and cost-efficient LLM serving for everyone.

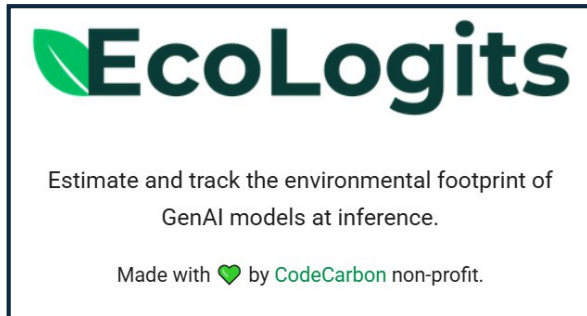
```
curl -fsSL https://ollama.com/install.sh | sh
```

Quantify the energy and environmental impact



Boundaries of our impact assessment methodology.

Quantify the energy and environmental impact



```
from ecologits import EcoLogits
from huggingface_hub import InferenceClient

# Initialize EcoLogits
EcoLogits.init(providers=['huggingface_hub'])

client = InferenceClient(model='meta-llama/Meta-Llama-3-8B-Instruct')
response = client.chat_completion(

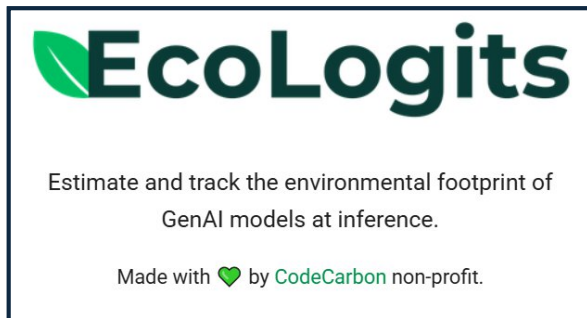
    messages=[{'role': 'user', 'content': 'Tell me a funny joke!'}]

)

# Get estimated environmental impacts of the inference
print(f'Energy consumption: {response.impacts.energy.value} kWh')
print(f'GHG emissions: {response.impacts.gwp.value} kgCO2eq')
```

Energy consumption: 3.850859137488289e-06 [3.764517004360749e-06 – 3.937201270615829e-06] kWh
GHG emissions: 1.720270322476616e-06 [1.6871537973155479e-06 – 1.753386847637684e-06] kgCO2eq

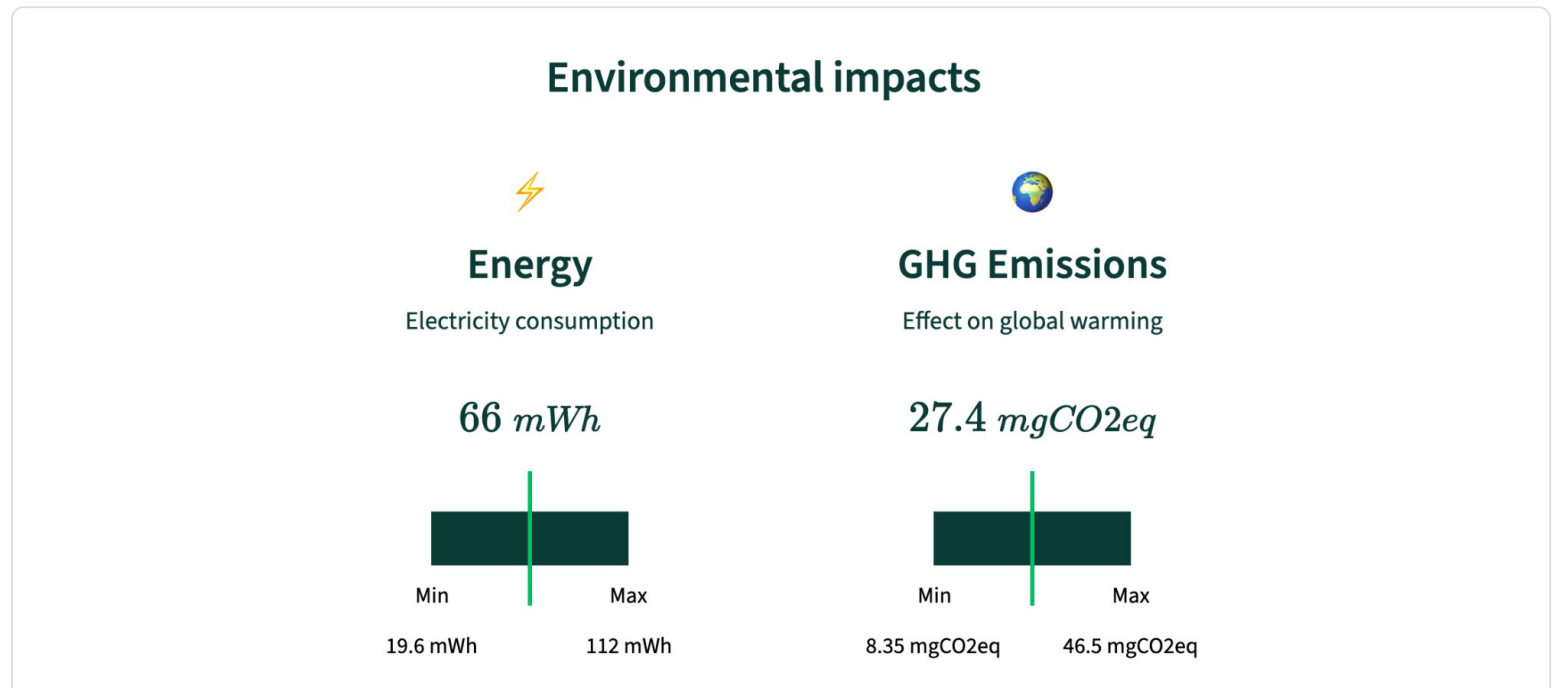
Quantify the energy and environmental impact



Reference : [EcoLogits Calculator](#), Hugging Face

Provider	Model	Example prompt
OpenAI	gpt 3.5 turbo	Write an article summary (250 out...

⚠️ The model architecture has not been publicly released, expect lower precision of estimations.



Quantify the energy and environmental impact



Estimate and track the environmental footprint of
GenAI models at inference.

Made with ❤️ by [CodeCarbon](#) non-profit.

Reference : [EcoLogits Calculator](#), Hugging Face

Configure the model

Provider

OpenAI

Model

gpt 3.5 turbo

Active parameters (B)

45

-

+

Total parameters (B)

45

-

+

Average TPS

53.10

-

+

Average TTFT (s)

0,42

-

+

Configure the prompt

Example prompt

Write a Tweet (50 output tokens)

Output completion tokens

50

-

+

Configure the data center

Data center PUE

1.20

-

+

Data center WUE [L / kWh]

0.60

-

+

Data center location

World

-

+

GHG emissions [kgCO₂eq / kWh]

0.473010

-

+

Abiotic resources [kgSbeq / kWh]

0.0000000134900

-

+

Primary energy [MJ / kWh]

2,587

-

+

Water consumption [L / kWh]

3,908

-

+



Reduce the impacts

True or False: *Grid Search* is the most energy-efficient way to find the best hyperparameters for a random forest model

`sklearn.grid_search.GridSearchCV`

```
class sklearn.grid_search.GridSearchCV(estimator, param_grid, scoring=None, loss_func=None, score_func=None,
fit_params=None, n_jobs=1, iid=True, refit=True, cv=None, verbose=0, pre_dispatch="2*n_jobs")
```

Exhaustive search over specified parameter values for an estimator.

Important members are fit, predict.

GridSearchCV implements a "fit" method and a "predict" method like any classifier except that the parameters classifier used to predict is optimized by cross-validation.



False

Reduce the impacts

For ML models

Hyperparameter tuning can consume a lot of energy due to the number of runs

Grid search



- Random search
- Bayesian optimization
- Hyperparameter optimization framework (Optuna)

Which technique is not a **POST**-training technique to have a more energy efficient model ?

Early stopping

Reduce the impacts

For DL models

Neural networks
without optimization



- Use pretrained models (transfer learning)
- Pruning
- Quantization
- Early stopping
- Efficient hyperparameter search

True or False: for an LLM, generating a long response is more energy-intensive than processing a long prompt

GPT-5.4

Our most capable model for professional work.

Price

Input:

\$2.50 / 1M tokens

Cached input:

\$0.25 / 1M tokens

Output:

\$15.00 / 1M tokens



True

Order these LLM factors from **Highest** to **Lowest** energy consumption impact for a single query

Bon ordre :

Number of parameters

Length of the answer

Length of the prompt

Temperature of the model

Reduce the impacts

For LLM models

Newest and largest LLM



- Right-size the model (smaller open models and distilled versions)
- Limit output length
- Optimize prompts
- Route intelligently (multi-model strategy)

Thank you for your attention

Questions?